UNIT 1 · TOPIC 1(B)

Farm Power Availability and Trends in Mechanization

Syllabus topic: Status of farm mechanization in India; power availability on farms and trends in mechanization

Farm power availability over time

Exam tip: memorize these figures.

Year	Farm power availability (kW/ha)
1951–52	~0.25
1975–76	0.48
2013–14	1.84
2018–19	2.49
2019–20	2.54
2021–22	~3.04
Target 2030	4.0

- Farm power availability (kW/ha) correlates strongly with food-grain productivity (t/ha) across Indian states. **Punjab and Haryana** have the highest power availability and the highest productivity.

Shift from animate to mechanical power

1. **1960–61:** animate sources (humans + draught animals) supplied **~93–97%** of farm power.
2. **2018–19 onward:** mechanical + electrical sources supply **>90%** of total farm power.

Power source	Approximate present share of farm power
Tractors	~47–50%
Electric motors	~25–27%
Diesel engines	~8–9%
Draught animals	~5%
Agricultural workers	~5%
Power tillers	~1%

The **tractor** is the single largest source of farm power in India.

Practice MCQs

Q1. The average farm power availability in India in 2021–22 was approximately:

(a) 1.84 kW/ha (b) 2.02 kW/ha
(c) 3.04 kW/ha (d) 4.0 kW/ha

Q2. The Government of India target for farm power availability by 2030 is:

(a) 3.5 kW/ha (b) 4.0 kW/ha

(c) 4.5 kW/ha (d) 5.0 kW/ha

Q3. The single largest source of farm power in India at present is:

(a) Electric motors (b) Diesel engines

(c) Tractors (d) Draught animals

Answers

Q1 — (c) 3.04 kW/ha

Q2 — (b) 4.0 kW/ha

Q3 — (c) Tractors